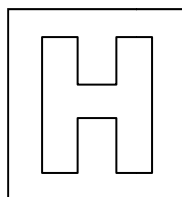


Candidate Name: _____

Class Adm No

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2014 Preliminary Examination II

Pre-University 3

H2 Biology

9648/01

Paper 1

23 September 2014

1 hour 15 min

Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

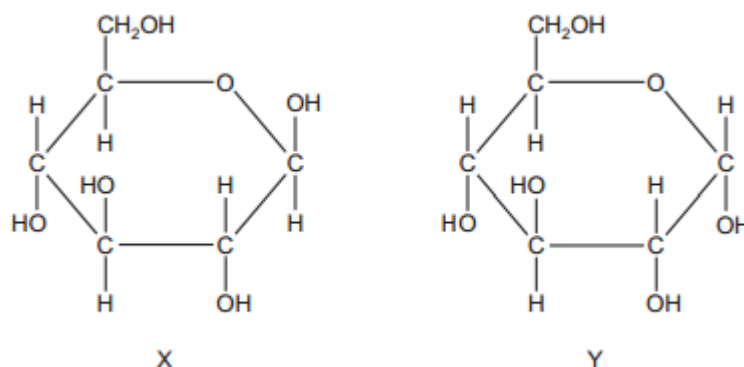
Calculators may be used.

This question paper consists of 22 printed pages.

[Turn over

Answer **all** questions.

1. The diagram below shows the structure of two hexose sugars, X and Y.



Which of the following statements about X and Y are correct?

- 1 X is found in amylose.
- 2 X is found in amylopectin.
- 3 Y is found in cellulose.
- 4 Y is found in glycogen.
- 5 Many residues of X can be bonded together by β (1 $\rightarrow$ 4) glycosidic bonds.

A 1 and 3 only

B 4 and 5 only

C 1, 2 and 3

D 2, 4 and 5

2. Acetylglucosamine is a naturally occurring polysaccharide. It is an unbranched chain in which monomers are linked by β (1 $\rightarrow$ 4) glycosidic bonds, with $-\text{CH}_2\text{OH}$ groups alternating on each side of the polysaccharide chain.

Which of the following is likely to be a property of acetylglucosamine?

A soluble in water due to $-\text{OH}$ groups projecting in all directions

B high tensile strength due to extensive cross-linking between neighbouring chains

C easily hydrolysed due to presence of β (1 $\rightarrow$ 4) glycosidic bonds

D compact due to helical structure with $-\text{OH}$ groups projecting to interior

3. The fluidity of cell membranes in eukaryotic cells can be regulated according to temperature by modifying the composition of the membrane in a number of ways.

Which of the following changes occur within the cell membrane at low temperatures?

- 1 increase in cholesterol content
- 2 decrease in cholesterol content
- 3 increase in unsaturated fatty acid chains
- 4 increase in saturated fatty acid chains
- 5 decrease in length of fatty acid chains
- 6 increase in length of fatty acid chains

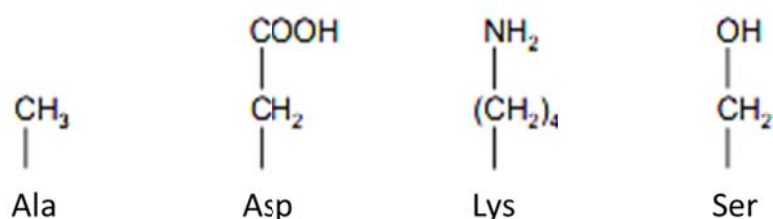
A 1, 3 and 5

B 1, 4 and 6

C 2, 3 and 5

D 2, 4 and 6

4. The diagram below shows the chemical structures of the R groups of 4 different amino acids. The R groups of amino acids are involved in forming various bonds and interactions in the tertiary structure of a protein.



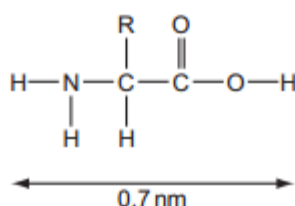
Which of the following correctly identifies the type of interaction that these different amino acids may be involved in?

	Ala	Asp	Lys	Ser
A	Hydrophobic	Ionic	Peptide	Hydrogen
B	Hydrogen	Disulphide	Ionic	Hydrophobic
C	Hydrogen	Disulphide	Peptide	Hydrophobic
D	Hydrophobic	Ionic	Ionic	Hydrogen

5. The table below shows the diameters of some atoms when they form single and double bonds.

atom	single bond / nm	double bond / nm
C	0.154	0.134
N	0.140	0.120
O	0.132	0.110
H	0.060	-

The approximate length of an amino acid was estimated using the diameters provided in the table above.



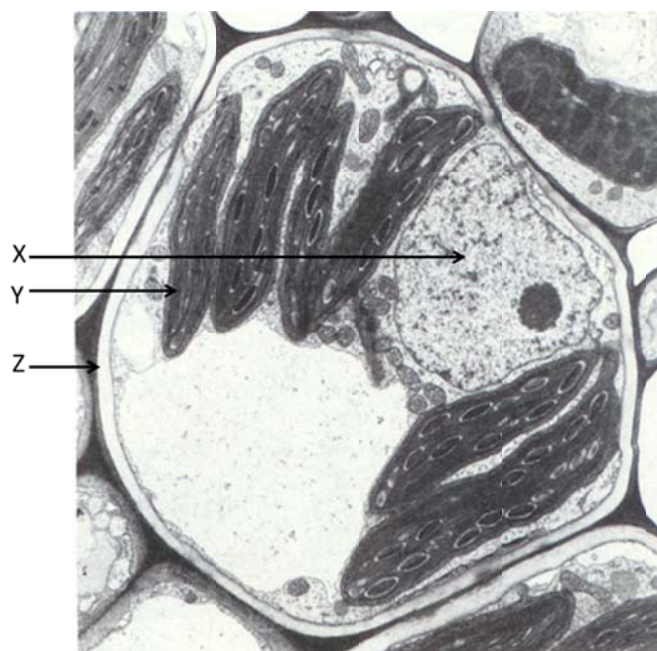
What would be the approximate length of a tripeptide formed using this amino acid?

- A 0.9 nm
- B 1.5 nm**
- C 1.7 nm
- D 2.1 nm
6. Lipase is an enzyme that is able to hydrolyse triglycerides. In an investigation to determine the rate of activity of the enzyme lipase on triglyceride hydrolysis, the pH of the reaction mixture had to be carefully maintained at an optimum to prevent denaturation of the enzyme.

Why was there a need to maintain the pH of the reaction mixture at its optimum?

- A The removal of water molecules from the solution during hydrolysis decreases the pH of the reaction mixture.
- B The products of hydrolysis of triglycerides decrease the pH of the reaction mixture.**
- C The products of hydrolysis of triglycerides increase the pH of the reaction mixture.
- D The addition of water molecules to the solution during hydrolysis increases the pH of the reaction mixture.

7. The diagram below shows an electron micrograph of a cell.



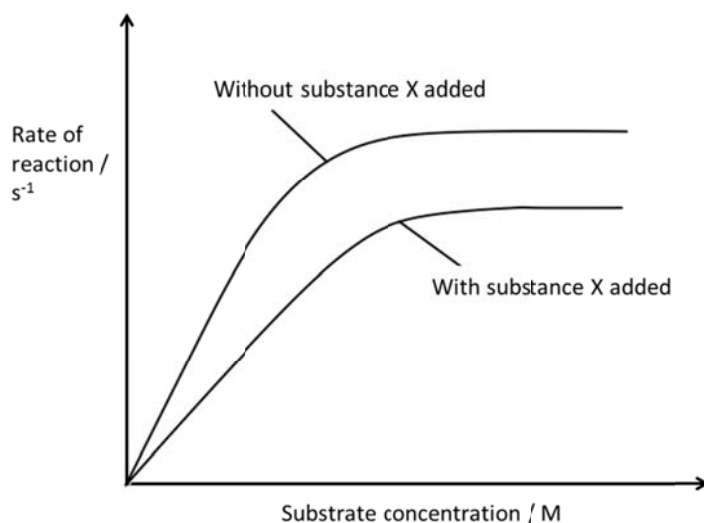
Which of the labelled components allows the cell to be identified as a eukaryotic cell?

- A X only
- B Z only
- C X and Y only
- D X, Y and Z
8. A student mixed up the labels of two solutions, 1 and 2. He knows that one contains starch and sucrose, and the other contains fat and galactose. In order to confirm the identities of the two solutions, a number of food tests were used.

Which of the following results correctly allows the student to identify the two solutions?

	ethanol emulsion test		Benedict's test		acid hydrolysis followed by Benedict's test		iodine test	
	1	2	1	2	1	2	1	2
A	-	+	+	-	-	+	+	-
B	+	-	+	-	-	+	-	+
C	+	-	-	+	+	-	-	+
D	-	+	-	+	+	-	-	+

9. The graph below shows the change in the rate of reaction of an enzyme with and without the addition of substance X.



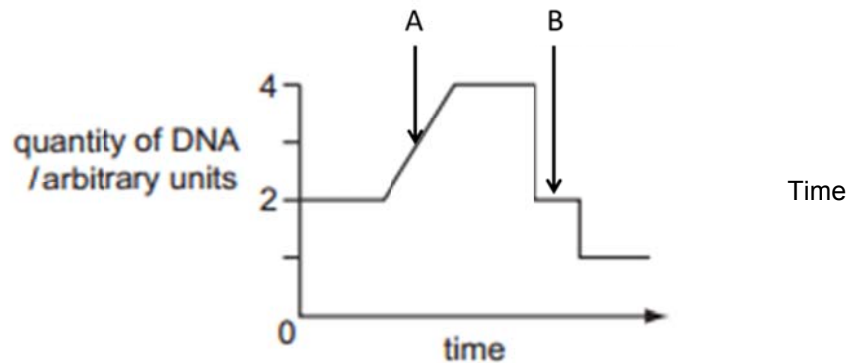
Which of the following statements about substance X is true?

- A The effect of substance X cannot be reduced by increasing the substrate concentration.
- B Substance X binds to the active site of the enzyme and competes with the substrate.
- C Substance X binds reversibly to the enzyme and changes the shape of its active site.
- D The effect of substance X can be reduced by increasing the enzyme concentration.
10. An enzyme has an optimum pH of 8. Under conditions of pH 5, enzyme denaturation can occur.

Which of the following could occur in the enzyme at pH 5?

- 1 acid hydrolysis of peptide bonds leading to loss of primary structure
 - 2 breakage of hydrogen bonds in the β -sheet secondary structure
 - 3 disruption of ionic bonds between acidic and basic R-groups of amino acids
 - 4 change in 3D shape of active site resulting in loss of function
- A 1 and 3 only
- B 3 and 4 only
- C 1, 2 and 4 only
- D All of the above

11. The graph below shows the changes in the quantity of DNA in a cell during meiosis. The cell undergoing nuclear division has a diploid number of $2n = 16$.



With reference to the figure, which of the following statements are true?

- 1 At point A, the chromosome number in the cell is increasing from 16 to 32.
- 2 At point A, the double stranded DNA helix unwinds due to action of DNA helicase.
- 3 At point B, the chromosome number in the cell is $n=8$.
- 4 At point B, homologous chromosomes pair up to form bivalents.

A 2 only

B 3 only

C 2 and 3 only

D All of the above

12. Gene mutations in tumour suppressor genes and proto-oncogenes can result in the development of cancer.

Which of the following statements about these two types of genes is true?

- A** A loss-of-function mutation in only one copy of a proto-oncogene is sufficient to induce excessive cell division.
- B** A gain-of-function mutation in both copies of a proto-oncogene has to occur before excessive cell division will be observed.
- C** A loss-of-function mutation in both copies of one tumour suppressor gene is sufficient to cause cancer.
- D** A gain-of function mutation in a proto-oncogene is unlikely to cause cancer if there are tumour suppressor genes are still functional.

13. Taxanes are complex terpenes produced by yew trees. An example of a taxane, paclitaxel, is used as a chemotherapy drug to treat breast cancer and ovarian cancer. Paclitaxel works by causing mitotic cells to stop dividing at metaphase.

Which of the following statements correctly identifies how paclitaxel may work?

- 1 It inhibits migration of the centrioles to opposite poles of the cell.
- 2 It prevents shortening of spindle fibres.
- 3 It inhibits condensation of chromatin in the nucleus.
- 4 It prevents homologous chromosomes from migrating to opposite poles.

A 1 only

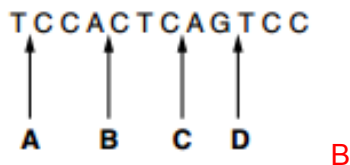
B 2 only

C 1 and 3 only

D 2 and 4 only

14. In a new species discovered in the Galapagos islands, the RNA triplet UAG acts as a stop codon in translation. The diagram below shows a short section of DNA from this species coding for four amino acids.

Where would a mutation, an addition of a thymine nucleotide, result in premature termination of transcription?



15. Which of the following statements about DNA replication is true?

A Okazaki fragments are joined together by DNA ligase, which catalyses the formation of phosphoester bonds.

B If there is a loss-of-function mutation in the gene coding for topoisomerase, DNA strands will be fragmented due to excessive supercoiling.

C The end replication problem is a result of DNA polymerase I being unable to fill the gap left by the primer at the 5' end of the leading strand.

D There is presence of a leading and lagging strand because DNA polymerase III can only function in one direction and add nucleotides to the 5' phosphate group.

16. What is the maximum number of hydrogen bonds that could be formed in a fragment of DNA that is 1200 nucleotides long?

A 400
B 1200
C 2400
D 3600

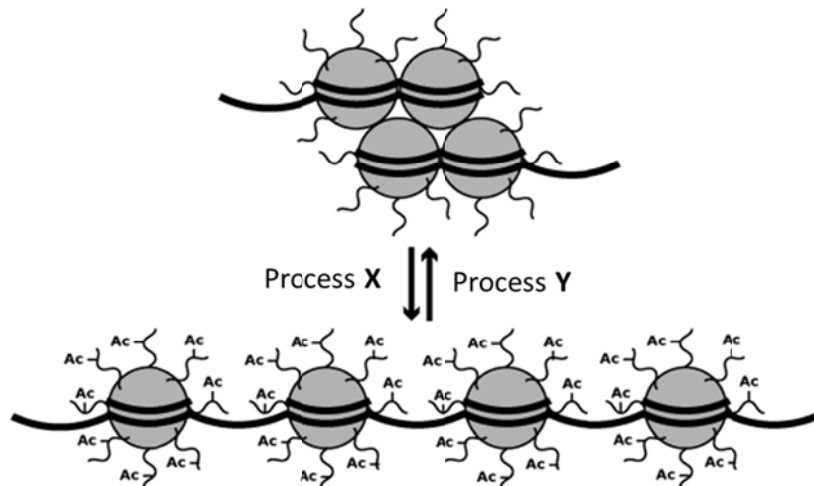
17. Post-transcriptional modifications are important in regulating eukaryotic gene expression. However, prokaryotes do not carry out post-transcriptional modifications.

Which of the following statements are true about post-transcriptional modifications?

- 1 Alternative splicing allows for the selective inclusion of certain introns into the mature mRNA, thus allowing one gene to code for more than one product.
- 2 Prokaryotes do not carry out post-transcriptional modifications because transcription and translation occur simultaneously.
- 3 Length of the 3' poly(A) is important for determining the half-life of the mRNA.
- 4 Presence of the 5' cap is required to facilitate transport of the mRNA from the nucleus to the cytoplasm.

A 2 and 4 only
B 1, 2 and 3 only
C 2, 3 and 4 only
D All of the above

18. The figure below shows two processes that can affect the rate of transcription in eukaryotes.

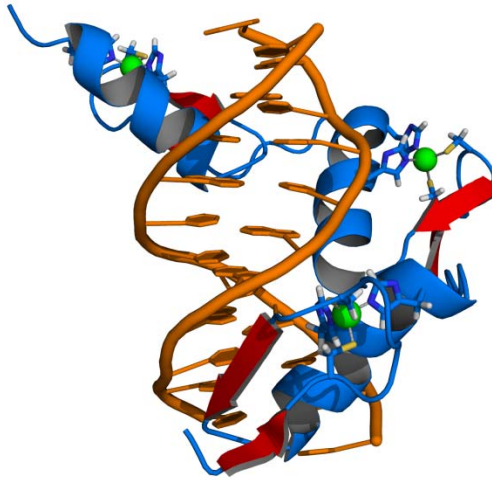


Which of the following statements regarding processes X and Y are incorrect?

- 1 Process X will be inhibited if leucine residues in histone tails are replaced with arginine residues.
- 2 Process X is facilitated by the enzyme histone acetyltransferase and is reversible.
- 3 Process Y prevents general transcription factors from accessing the operator region of genes, thus inhibiting transcription.
- 4 Process Y is facilitated by the enzyme histone deacetylase and is permanent.

- A** 1 and 3 only
- B** 2 and 4 only
- C** 1, 2 and 3 only
- D** 1, 3 and 4 only

19. The diagram below shows a zinc finger transcription factor binding to a region of DNA. The transcription factor has two domains, a DNA binding domain and an activation domain.

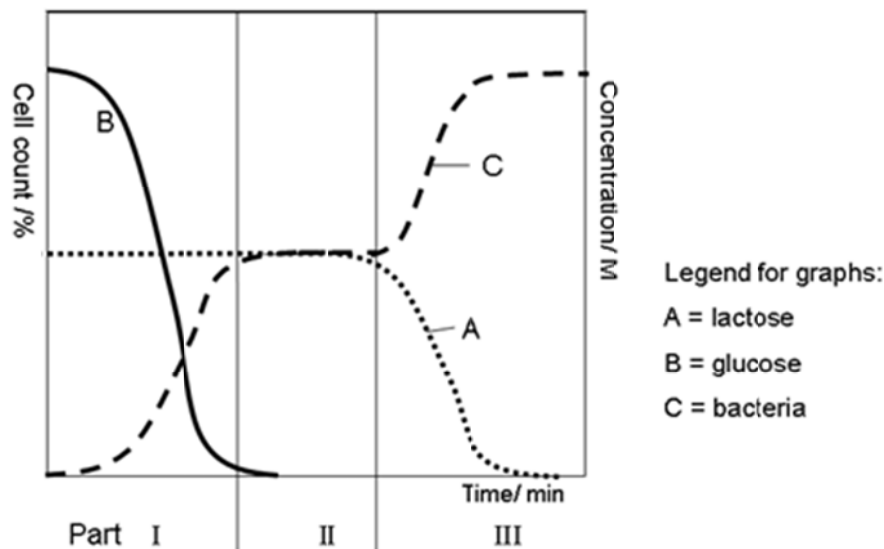


A mutation in the gene coding for the zinc finger protein results in an abnormal increase in the rate of transcription of a particular gene X.

Which of the following explains the outcome of the mutation?

- A The mutation affects the 3D shape of the activation domain, increasing its efficiency in recruiting DNA methyltransferase.
 - B The mutation affects the 3D shape of the activation domain, stabilising the interaction between the zinc finger protein and mediator proteins.
 - C The mutation affects the 3D shape of the DNA binding domain, stabilising the interaction between the zinc finger protein and the silence region.
 - D The mutation affects the 3D shape of the DNA binding domain, increasing its efficiency in recruiting histone acetyltransferase.
20. Which of the following does not increase genetic variation in a prokaryotic population?
- A Binary fission
 - B Transduction
 - C Transformation
 - D Conjugation

21. The graph below shows the number of bacteria growing in the presence of varying concentrations of glucose and lactose.



Which of the following statements correctly accounts for the bacteria growth at different parts of the graph?

- 1 In Part I, the *lac* repressor is not bound to the operator and the catabolite activator protein is not bound to the CAP-binding site, hence the *lac* operon is inactive.
- 2 In Part II, cAMP levels are high and binding of cAMP to the *lac* repressor inactivates the repressor, switching on transcription of the *lac* operon.
- 3 In Part III, allolactose levels are depleting, eventually resulting in activation of the *lac* repressor and a plateau in bacteria growth.

A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

22. The table below shows the difference in activity of various bacterial enzymes in the presence and absence of tryptophan for a normal cell and a mutant cell. Enzymes A, B and C are coded for by structural genes found in the *trp* operon.

Enzyme	Activity level of enzymes / arbitrary units			
	Normal		Mutant	
	Trp absent	Trp present	Trp absent	Trp present
A	800	0	800	800
B	800	0	800	800
C	800	0	800	800

Which of the following mutations could account for the phenotype of the mutant cell?

- A Loss-of-function mutation in the gene coding for enzyme A
 - B Loss-of-function mutation in the *trpR* gene**
 - C Gain-of-function mutation in the Trp activator
 - D Gain-of-function mutation in the operator sequence
23. There are three different types of influenza viruses, types A, B and C. A common vaccine for influenza targets all three of these viral strains, containing inactivated forms of all three viral strains.

However, the vaccine is usually effective only for one year, after which it loses its effectiveness and has to be reformulated. As a result, a new vaccine needs to be administered to an individual every year.

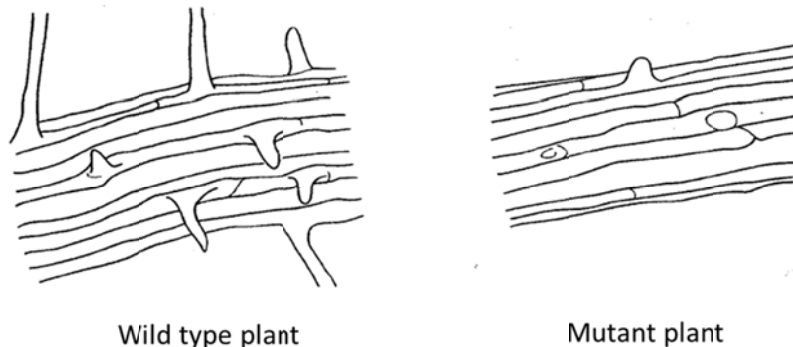
Why do influenza vaccines need to be reformulated every year?

- A The influenza genome consists of separate RNA segments which allow recombination between different haemagglutinin and neuraminidase genes.**
- B Antigenic drift may occur where different influenza strains infect the same host cell and exchange genetic material.
- C Antigenic shift may occur where random mutations result in change in structure of glycoproteins on the viral envelope.
- D Neuraminidase on the viral envelope may undergo mutation, resulting in a change in shape of the glycoprotein.

24. Azidothymidine (AZT) is used as a drug for the treatment of HIV. The chemical structure of AZT is similar to a nucleotide, however it does not have a 3' –OH group.

Which of the following could explain the mechanism by which AZT inhibits HIV reproduction?

- A AZT prevents the binding of HIV glycoproteins to cell surface receptors of T-helper cells, inhibiting the recognition of the host cell.
 - B AZT prevents integration of the HIV genome into the host DNA, allowing the immune system to remove infected cells from the body.
 - C AZT inhibits the activity of reverse transcriptase as it prevents elongation of viral DNA using its RNA genome as a template.
 - D AZT results in adhesion of new viral particles to glycoproteins on surface of the host cell, thus inhibiting the budding phase of new viruses.
25. A mutant form of the plant *Arabidopsis thaliana* was discovered to have very short root hairs. The mutant contains a mutation in the gene coding for the enzyme reduced NADP oxidase. The normal enzyme converts reduced NADP to NADP^+ .



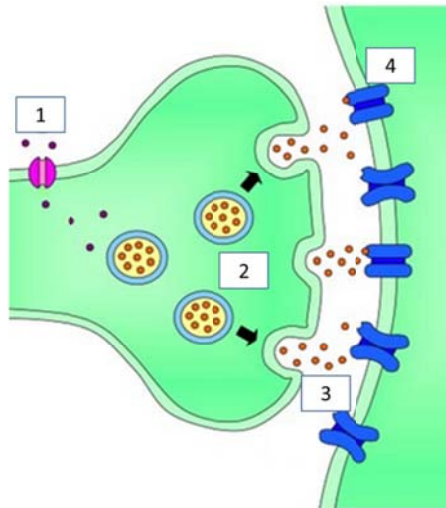
Which of the following could account for the presence of very short root hairs in the mutant?

- A As NADPH cannot be oxidised, rate of light-dependent reactions will decrease and cellular growth will also decrease.
- B As NADPH cannot be oxidised, rate of light-dependent reactions will decrease and less water is needed for photolysis, hence root hairs can be shorter.
- C As NADP^+ cannot be reduced, rate of light-independent reactions will decrease and cellular growth will also decrease.
- D As NADP^+ cannot be reduced, rate of light-independent reactions will decrease and less water is needed for photolysis, hence root hairs can be shorter.

26. Oligomycin is an inhibitor of ATP synthase, blocking the proton channel to prevent H^+ diffusion across the inner mitochondrial membrane.

Which of the following is not likely to be an effect of oligomycin?

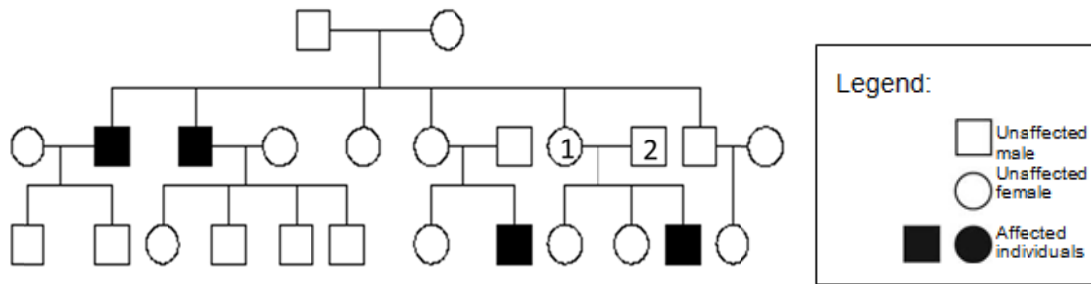
- A Inhibition of the electron transport chain
 - B Increase in lactic acid fermentation
 - C Inhibition of glycolysis
 - D Inhibition of the Krebs cycle
27. The figure below shows the transport of various molecules that takes place at a synapse.



Which of the following statements correctly describes transport of different molecules at the synapse?

- 1 The entry of Ca^{2+} ions into the pre-synaptic knob at location 1 takes place requires expenditure of energy
 - 2 Neurotransmitter molecules are transported out of the pre-synaptic knob via exocytosis at location 2.
 - 3 Neurotransmitter molecules are transported across the synaptic cleft via simple diffusion at location 3.
 - 4 Ions move across the membrane via ligand-gated channels at location 4 because they are large.
- A 1 and 2 only
 - B 2 and 3 only
 - C 1, 2 and 3 only
 - D All of the above

28. The pedigree diagram below shows the inheritance of haemophilia in a family. Haemophilia is a disease that usually affects males and results in the inability of blood clotting.



What is the probability that the next child of individuals 1 and 2 is a girl who is a carrier of haemophilia?

- A $1/2$
- B $1/4$
- C $1/8$
- D $1/16$
29. In an investigation into the maize genome, pollen grains from a pure-bred plant with dark green leaves and red corn cobs were transferred to stigmas of a pure-bred plant with light green leaves and yellow corn cobs. All the F_1 generation had dark green leaves and yellow corn cobs.

When plants of the F_1 generation underwent a test cross, the observed phenotypes of 200 offspring are as follows:

Dark green leaves, yellow corn cobs	Dark green leaves, red corn cobs	Light green leaves, yellow corn cobs	Light green leaves, red corn cobs
46	50	53	51

Which of the following statements about the test cross is correct?

- A The expected ratio is approximately equivalent to the observed ratio, therefore the genes for leaf colour and corn cob colour undergo independent assortment.
- B The expected ratio is approximately equivalent to the observed ratio, therefore the genes for leaf colour and corn cob colour have a recombination frequency of 0.5.
- C The expected ratio is not equivalent to the observed ratio, therefore the genes for leaf colour and corn cob colour undergo independent assortment.
- D The expected ratio is not equivalent to the observed ratio, therefore the genes for leaf colour and corn cob colour have a recombination frequency of 0.5.

30. The diagram below shows a section of the mitochondrial DNA sequence from different sub-species of pigeons found in different parts of the world. Dashes indicate that the bases are identical to the sequence from the United States.

United States	AAAGCAGAAGAAATAGATGCAAAGAAGAAGAGTTCAA
United Kingdom	--GCAT-----C-----AATT
Brazil	-----C-----TA-----
New Zealand	GG-CG-C-----GGG-GG-----CGGG
China	TTT--T-----TT--T-----GGTT
Mexico	-----C-----T-----
Russia	TTT--T-----TC--T-----GGTT
Chile	-----C-----C-TA-----
Ireland	--GCAT-----CC-----A-TT
Canada	-----C-----



Region A

Which of the following conclusions cannot be derived from the sequences above?

- A Pigeons from China and Russia shared a more recent common ancestor than pigeons from the United Kingdom and Ireland.
- B The sequence in Region A most likely codes for an important structure in the mitochondrion.
- C Pigeons from New Zealand have the most number of nucleotide differences from pigeons in the United States because of geographical isolation.**
- D Pigeons from Canada are likely to share more anatomical homologies with pigeons from United States compared to pigeons from Ireland.
31. The hooded crow and the carrion crow inhabit different regions of Europe, with the hooded crow being found in Northern and Eastern Europe while the carrion crow is found in Western Europe. There is an overlap of both types of crow in Germany and Denmark.

At regions of overlap, hybrids between the hooded crow and carrion crow can be found. However, these hybrids are not as well-adapted as pure bred birds and usually die before maturity.

Which of the following statements about the hooded crow and carrion crow is likely to be true?

- A Allopatric speciation occurs due to both types inhabiting different geographical regions of the European continent.
- B Sympatric speciation occurs due to hybrids of both types of crow having decreased hybrid vigour.**
- C Allopatric speciation occurs due to presence of a physical barrier between populations of hooded crows and carrion crows.
- D Sympatric speciation occurs due to assortative mating by both types of crow.

32. Which of the following statements about Neo-Darwinism is not accurate?

- 1 Variations within a population are due to spontaneous mutations only.
- 2 Individuals with genetic variations best suited to the environment will survive till maturity to reproduce.
- 3 Differential reproductive success will lead to changes in allele frequencies over time.
- 4 Speciation will only occur if populations of the same species are unable to interbreed.

A 1 only

B 1 and 2 only

C 2 and 4 only

D 1, 2 and 3 only

33. A scientist conducted an experiment on two separate cultures of liver cells from the same sheep in order to study the expression of Glut4 glucose transporters on the surface of liver cells. In this experiment, the glucose concentration in the external culture media was measured at fixed time intervals to determine uptake of glucose into the cells.

Cultures A and B were both exposed to saline containing a fixed volume of insulin. Cells in the control experiment were exposed to saline only. The results of the experiment are shown in the table below.

Time / min	Glucose level in culture medium / mg dL ⁻¹		
	Control	A	B
0	400	400	400
15	382	342	379
30	370	230	340
45	366	100	301
60	359	85	270

Which of the following best explains the difference in results obtained for cultures A and B?

A Cells in culture B had fewer number of insulin receptors on the cell surface compared to culture A.

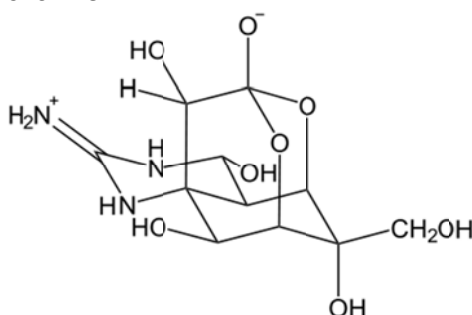
B Cells in culture B had previously been exposed to high amounts of insulin, resulting in desensitisation of cells to insulin.

C Cells in culture B were contaminated with trace amounts of glucagon.

D Cells in culture B were kept at higher temperatures than cells in culture A, resulting in them having a higher rate of respiration.

34. Which of the following correctly describes the action of a trimeric G-protein?
- A G-protein is phosphorylated and activated by receptor tyrosine kinase.
 - B Presence of an intracellular signal molecule activates the G-protein coupled receptor, leading to activation of G-protein.
 - C G-protein has GTPase activity, hydrolysing GDP to GTP.
 - D Activated subunits of the G-protein travel along the cell surface membrane to activate adenylyl cyclase.**

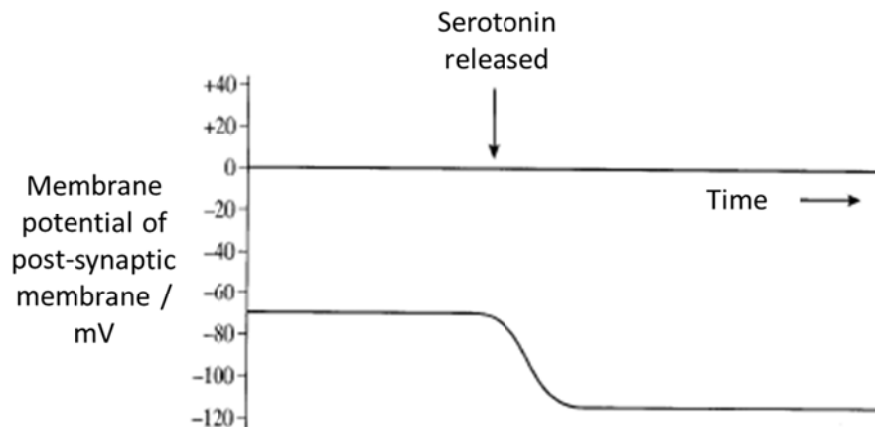
35. The diagram below shows the molecular structure of tetrodotoxin, a potent neurotoxin that is found in pufferfish and porcupinefish. Tetrodotoxin binds at the pore opening of the voltage-gated Na^+ channel.



Which of the following explains why tetrodotoxin is a lethal toxin?

- A Tetrodotoxin prevents the closing of the voltage-gated Na^+ channel, thus extending the depolarisation phase of the action potential.
- B Tetrodotoxin prevents the closing of the voltage-gated Na^+ channel, thus extending the repolarisation phase of the action potential.
- C Tetrodotoxin prevents the opening of the voltage-gated Na^+ channel, thus inhibiting the depolarisation phase of the action potential.**
- D Tetrodotoxin prevents the opening of the voltage-gated Na^+ channel, thus inhibiting the repolarisation phase of the action potential.

36. Serotonin is an inhibitory neurotransmitter that functions to regulate appetite and sleep. The graph below shows the effect of serotonin release on the membrane potential of the post-synaptic membrane.



Which one of the following is a possible effect of serotonin on the post-synaptic membrane?

- 1 opening of Cl^- channel to allow influx of Cl^-
- 2 opening of Cl^- channel to allow efflux of Cl^-
- 3 opening of K^+ channel to allow influx of K^+
- 4 opening of K^+ channel to allow efflux of K^+

A 1 and 3 only

B 1 and 4 only

C 2 and 3 only

D 2 and 4 only

37. Plasmids are vectors that are commonly used for cloning of genes.

Which of the following features of plasmids do not contribute towards its role in gene cloning?

A Origin of replication

B Circular chromosome made of DNA

C Presence of genetic markers

D Multiple cloning site

38. Induced pluripotent stem (iPS) cells are obtained by reprogramming gene expression in somatic cells. The method of reprogramming often involves the use of viruses. This method of obtaining pluripotent stem cells circumvents the ethical issue of obtaining these stem cells from embryos. Despite the advantages of iPS cells, there are still some problems with its use in medical treatment.

Which of the following is not a valid argument against the use of iPS cells?

- A iPS cells can only be used for the treatment of single gene disorders.
- B There is a danger of iPS cells undergoing excessive cell division after transplantation, leading to formation of tumours.
- C Reprogrammed iPS cells do not exhibit identical gene expression patterns as normal embryonic stem cells.
- D Use of retroviruses in generation of iPS cells poses the risk of viruses integrating its genome into inappropriate locations within the host cell genome.
39. The Human Genome Project was started in the 1990s to sequence the entire human genome. There were several project goals that were identified prior to the start of the project.

Which of the following are goals of the Human Genome Project?

- 1 to determine the sequence of the 3 billion base pairs making up the human genome
 - 2 to improve tools used for analysing human genome data
 - 3 to identify genes involved in disease in order to improve molecular diagnosis
 - 4 to address ethical, legal and social issues that may arise from the project
 - 5 to determine the relative distances between different genes on chromosomes in order to create a linkage map
- A 1, 2 and 3 only
- B 1, 2 and 4 only
- C 1, 3 and 4 only
- D All of the above

40. Why are plant cells and prokaryotic cells more commonly used for genetic engineering than animal cells?
- A Plant cells and prokaryotic cells do not contain introns, hence there is no requirement for splicing.
 - B Somatic plant cells can undergo tissue cloning to produce new plantlets.**
 - C Plant cells and prokaryotic cells have smaller genomes than animal cells.
 - D Vectors used for genetic engineering have a higher probability of destroying animal cells due to the lack of cell wall.